Customer Churn ANN — Statistical Evaluation Report

Generated: 2026-10-29T13:10:44.212719-04:00

1. Executive summary

The model was evaluated on **2,000 held-out test observations**. Accuracy was **85.90%**, ROC-AUC was **0.8582**, precision was **78.03%**, and recall was **42.75%** at a classification threshold of **0.50**.

The ROC-AUC indicates strong ranking discrimination on this test set. Recall is relatively low at the selected threshold. Review threshold tuning on validation data if identifying more churners is a business priority.

These results describe performance on this test set; they do not by themselves establish statistical significance or business impact.

2. Evaluation metrics

Metric	Result
Accuracy	0.8590 (85.90%)
Precision	0.7803 (78.03%)
Recall / Sensitivity	0.4275 (42.75%)
F1 score	0.5524
ROC-AUC	0.8582
Average precision (PR-AUC field)	0.6969
Specificity	0.9692 (96.92%)
Classification threshold	0.50

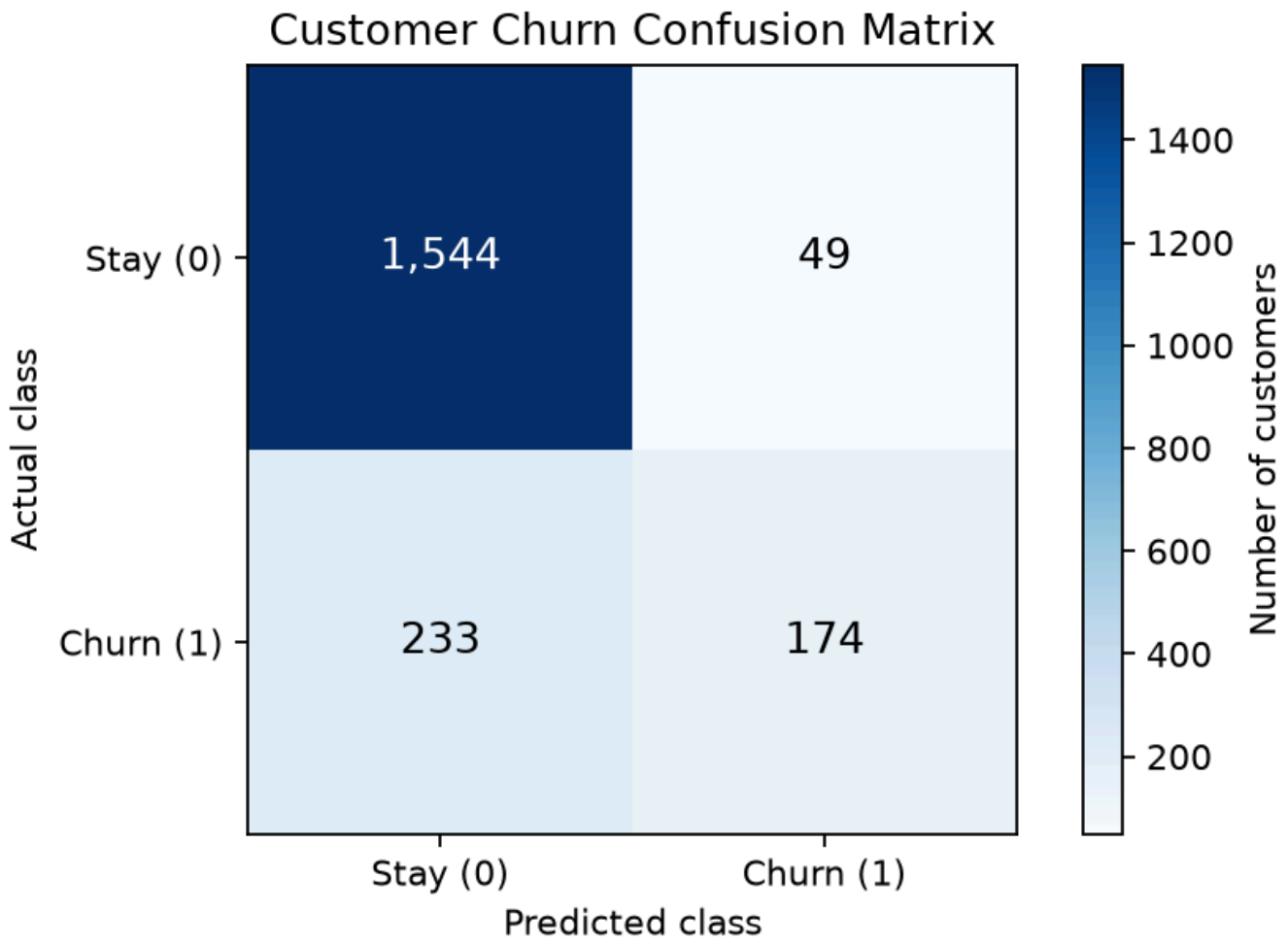
Average precision is calculated with scikit-learn's `average_precision_score`; it is not the trapezoidal integral of the precision-recall curve.

3. Confusion matrix

Actual class	Predicted stay (0)	Predicted churn (1)	Total
Stay (0)	1,544	49	1,593
Churn (1)	233	174	407
Total	**1,777**	**223**	**2,000**

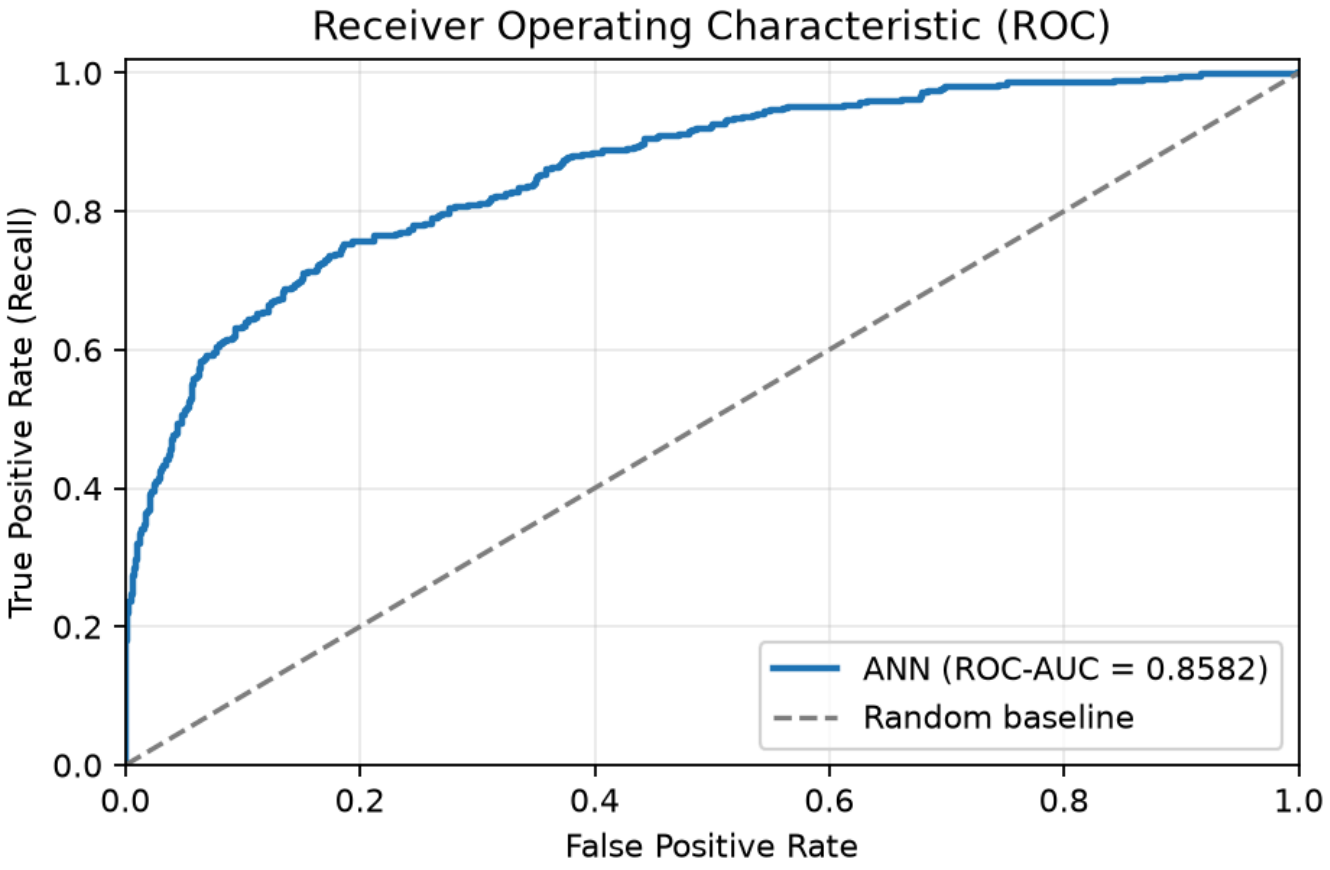
- False-positive rate: **3.08%**.
- False-negative rate: **57.25%**.
- Churn prevalence in test set: **20.35%**.

At this threshold, the model identified 174 of 407 actual churners and missed 233. It flagged 223 customers as potential churners.

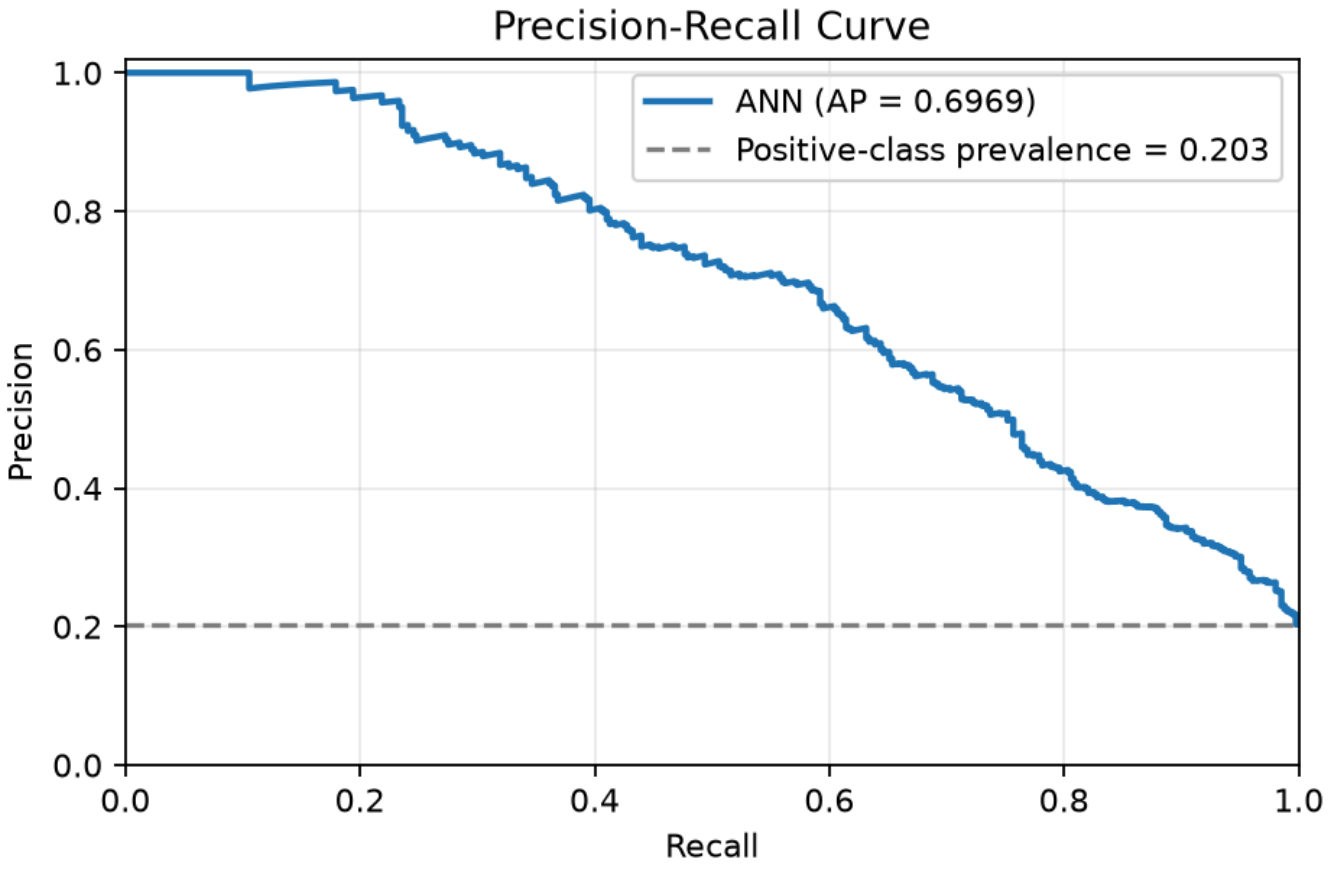


4. Diagnostic charts

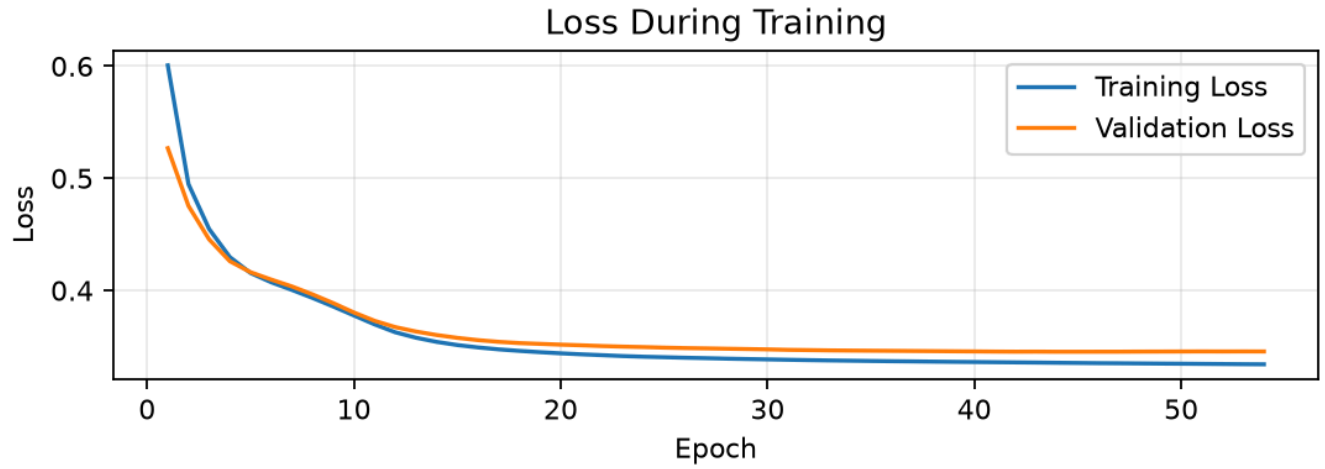
ROC curve

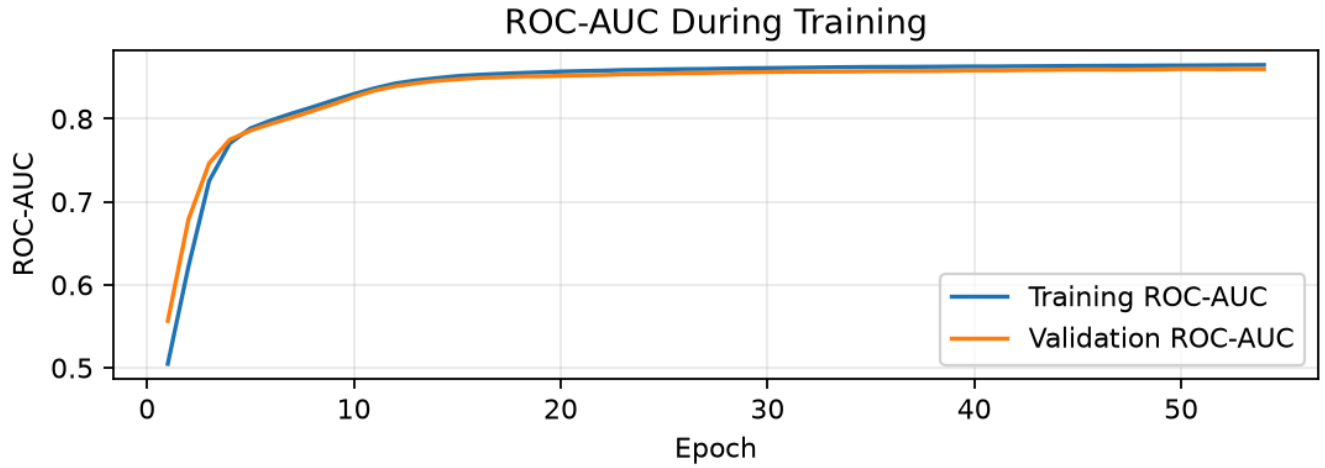


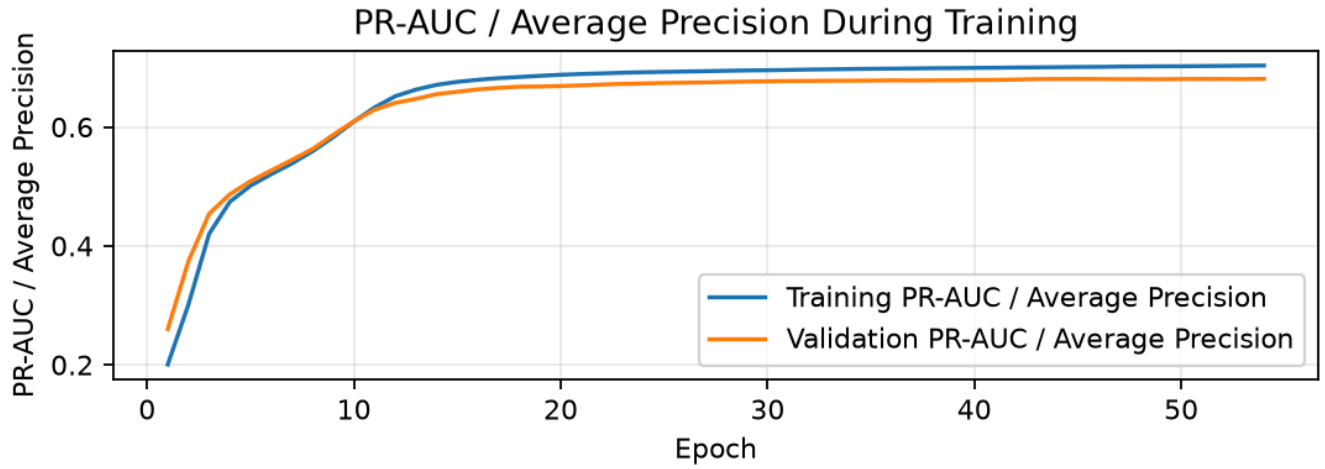
Precision-recall curve

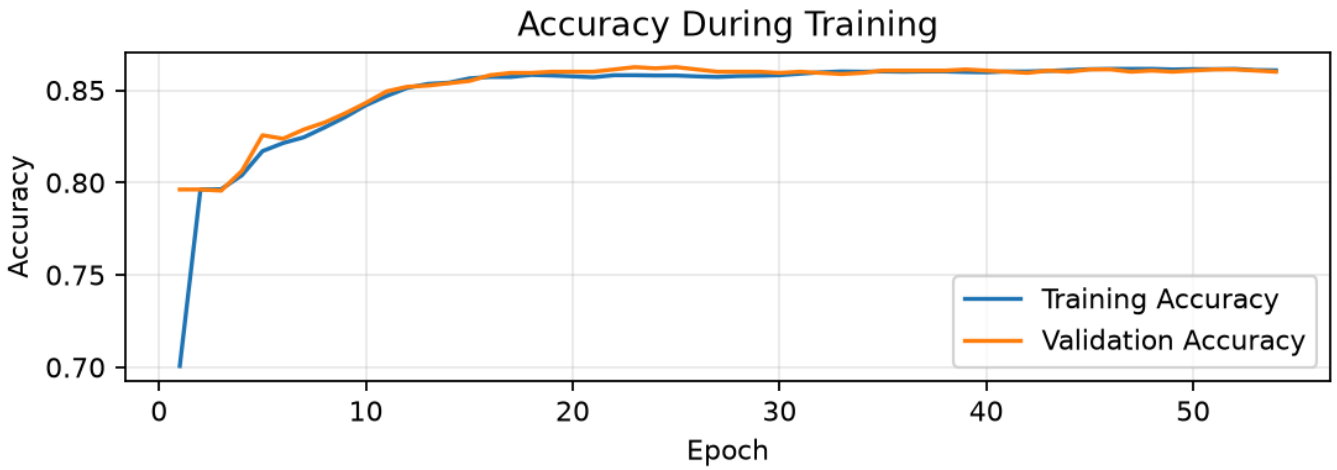


Training and validation curves


Loss During Training
Training Loss
Validation Loss
Loss
Epoch
0.6
0.5
0.4
0
10
20
30
40
50


ROC-AUC During Training
Training ROC-AUC
Validation ROC-AUC
ROC-AUC
Epoch
0.8
0.7
0.6
0.5
0
10
20
30
40
50


PR-AUC / Average Precision During Training
Training PR-AUC / Average Precision
Validation PR-AUC / Average Precision
PR-AUC / Average Precision
Epoch
0.6
0.4
0.2
0
10
20
30
40
50


Accuracy During Training
Training Accuracy
Validation Accuracy
Accuracy
Epoch
0.85
0.80
0.75
0.70
0
10
20
30
40
50

Review the curves for divergence between training and validation performance. Early stopping alone does not prove that overfitting has been eliminated.

5. Interpretation and limitations

- Precision and recall reflect the selected classification threshold.
- ROC-AUC measures ranking discrimination, not probability calibration.
- Average precision should be interpreted relative to churn prevalence and suitable baseline models.
- The results do not establish causality or the effectiveness of retention interventions.
- Confirm that preprocessing was fitted only on training data and that the test set was not used for model or threshold selection.

6. Recommendations

1. Tune the classification threshold using validation data; do not select it using the final test set.
2. Compare against logistic regression and tree-based baselines using identical data splits.
3. Estimate intervention costs and retention benefits before selecting an operating threshold.
4. Assess probability calibration and evaluate performance on later customer cohorts.

7. Reproducibility

- Test observations: 2,000
- Positive-class prevalence: 0.2035
- Decision threshold: 0.50
- Detailed metrics: `metrics.json`
- Model and preprocessing artifacts should be versioned separately.

This report was generated automatically from the current run's test labels, predicted probabilities, evaluation metrics, and training history.